



Monetisation of digital health data through a GDPR-compliant and blockchain enabled digital health data marketplace: A proposal to enhance patient's engagement with health data repositories[☆]

Mohamed Maher^{a,b,*}, Imtiaz Khan^b, Verma Prikshat^c

^a Balsamee LTD, Regus House, Falcon Drive, Cardiff CF10 4RU, UK

^b Cardiff School of Technologies, Cardiff Metropolitan University, Llandaff Campus, Cardiff CF5 2YB, UK

^c Cardiff School of Management, Cardiff Metropolitan University, Llandaff Campus, Cardiff CF5 2YB, UK

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ABSTRACT

With the growing use of digital technologies, the vision of a predictive, preventive and personalised healthcare system, commonly termed healthcare 4.0, is becoming tangible. However, the manifestation of healthcare 4.0 relies on empowering patients as active stakeholders of the healthcare system and transforming traditionally siloed data repositories into accessible ones. Combining blockchain-like distributed technology and Airbnb-like sharing economy model, this paper proposes a digital health data marketplace (DHDM) through which patients, as the data producer, can sell their health data to data consumers like medical practitioners, researchers, policymakers and machine learning algorithm developers without compromising privacy and security. In conjunction with the Welsh Secure Anonymised Information Linkage (SAIL) Databank, a traditional centralised and siloed data repository, DHDM was evaluated in a simulated environment to understand the technical feasibility, regulation compliance and, most importantly, the impact of such marketplace on healthcare 4.0 transformation.

1. Introduction

Recent years have witnessed the growing use of electronic health records (EHR) in healthcare settings as well as Healthcare Wearable Devices (HWDs) like smartwatches or Fitbit in normal settings. As a result, a large volume and variety of health data are being produced, creating the potential to transform health care from conventional reactive medicine to predictive diagnosis along with targeted preventive measures and personalised treatment regimens. A UK government-commissioned review found that data from the National Health Service (NHS) could transform the healthcare industry and lead to the emergence of AI-based diagnostics (NHS, 2019; Harwich, 2018; NHS, 2022). A project like DeepMind Health is a recent example of this perspective. The AI company based in London and owned by Alphabet developed the mobile app Streams (Royal-Free, 2016; Powles & Hodson, 2017) that uses London Royal Free hospital's EHR data to predict and identify patients who are subjected to acute kidney injury, a condition linked to 100,000 deaths in the UK every year (Kollewe, 2021). Also, portals like PatientsLikeMe (Patientslikeme, 2021) through which patients with similar medical conditions and/or concerns can share information regard-

ing their treatments, and vital parameters, have demonstrable benefits to their users (Frost & Massagli, 2008; Wicks et al., 2010).

Despite these potential and demonstrable impacts, the Predictive, Preventive and Personalised Medicine (PPPM) aspiration outlined by healthcare 4.0 remains intangible (Golubnitschaja et al., 2014). This is due to the unavailability of the right health data at the right time to data consumers, i.e. the researchers, medical practitioners, machine learning algorithms and policymakers. Accessibility of data remains a crucial bottleneck (Bezuidenhout & Shanahan, 2022) despite initiatives like the FAIR (Findable, Accessible, Interoperable, and Reusable) data principle (Wilkinson et al., 2016; Wilkinson et al., 2019) in health data stewardship practice (Inau et al., 2021) and the implementation of GDPR (GDPR, 2016) that fundamentally changed the ownership and stewardship of health data from the service provider to the patient and stated the legislative rights to access to individual's own data (GDPR, 2018). Having GDPR in practice, it is now more evident that the data has a single owner and gatekeeper regarding accession or distribution rights (Rowell & Sebro, 2022). Even anonymously, only the patient has the right to grant access to their data and allow using the information in a way that will benefit everybody. Ironically, the patients have not yet

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* Corresponding author at: Balsamee LTD, Regus House, Falcon Drive, Cardiff CF10 4RU, UK.

E-mail address: mmaher@cardiffmet.ac.uk (M. Maher).

recognised the value of this new ownership right and how to manage the stewardship. This is because no direct wins are created to serve them, and indirect wins are not clear enough to overcome the legitimate worries of data leakage and subsequent privacy exposure. Therefore the target is to find a novel approach and tools to balance individual privacy and transparent data access for research purposes (Kostkova et al., 2016).

This paradigm shift introduced by GDPR to the service providers also brought opportunities for individuals to monetise their medical data by selling data to medical researchers or technology companies. Similar to Airbnb, which enabled individuals to monetise their spare accommodations, patients, with their new ownership and other rights bestowed by GDPR can sell their medical data at their discretion. However, with the current centralised data management framework where EHRs are fragmented across different service providers and where regulations differ across organisations and geographical jurisdictions, access and stewardship will be challenging, especially the microtransactions in such a distributed environment. In this context, Blockchain like DLT technologies and associated smart contracts, have been considered a game-changer with an inbuilt distributed architecture and the ability to administer information governance in a decentralised manner for diverse types of transaction-based digital services.

Distributed Ledger Technology (DLT), commonly known as Blockchain or Web 3.0, was introduced just after the 2008 financial crisis by a pseudonymous author Satoshi Nakamoto, as the backbone of the cryptocurrency Bitcoin (Nakamoto, 2008; Abramova and Bo'hme, 2016; Cocco and Marchesi, 2016). Blockchain technology is a digital system for recording the transaction of assets or data through a democratic consensus-based decentralised system where a network of computers across the globe participate as nodes and record the validated transaction data in a synchronised and immutable manner.

Recent Blockchain research has focused mainly on FinTech (Ali et al., 2020), becoming the new norm in financial services. However, the potential of Blockchain in securing and bookkeeping medical data transactions is as high as other FinTech applications (Mettler, 2016). Blockchain will promote new digital business models and digital health initiatives. This technology and its applications open new doors concerning how market interactions in healthcare can be done. Blockchain thus has immense potential for the future and will show disruptive changes in the healthcare industry (Mettler, 2016; Dimitrov, 2019; Tandon et al., 2020; Przhedetskiy et al., 2019). Blockchain has been applied in a variety of health and economics areas ranging from FinTech to internet companies for value co-creation (Ma'culiene & Skar'auskiene, 2021); cyber security (Mittal et al., 2021); privacy in digital analytics (Gupta et al., 2020); businesses (Grover et al., 2018) and different aspects of shared and circular economy (Kouhizadeh et al., 2020).

Inspired by the sharing economy model of Airbnb and taking advantage of the paradigm shift introduced by the GDPR in the context of policy and Blockchain in the context of data management, privacy and security, here we propose the conceptual design of a 'Digital Health Data Marketplace (DHDM)' model. The marketplace will allow patients to contribute and manage their data with various data consumers, gain financial rewards, or buy digital health services without disclosing their identity. As a case study, we have evaluated the marketplace in conjunction with a traditional health databank - Welsh Secure Anonymised Information Linkage (SAIL) Databank (Secure Anonymised Information Linkage Databank, 2019) to understand the marketplace's technical, legislative and financial feasibility.

The rest of the paper is set as follows; Section 2 introduces the DHDM in detail, along with the stakeholders involved. Section 3 briefly describes the SAIL databank and its workflow in relation to data access and GDPR. Section 4 introduces a matrix to analyse the SAIL databank workflow in its current state and a case study evaluating the SAIL databank in conjunction with the DHDM. The paper concluded in Section 5, outlining the sociotechnical limitations of the proposed marketplace with recommendations for future work.

2. Digital health data marketplace (DHDM)

Beyond FinTech, one of the prospering trends in Blockchain technology is the applications related to patient health record security and the safe exchange of information between caregiving organisations (Leeming et al., 2019; Hasselgren et al., 2020). Estonia's Blockchain-backed digital health infrastructure is a perfect example of largescale implementations of such applications.

2.1. The DHDM data monetisation approach

In this article, the authors propose a model for a Blockchain-based trusted marketplace solution to put the needed dynamics in place and tools in the hand of the patient to control the access to their personal health data and release the resources to support the much-needed healthcare innovation. This model builds on Blockchain success in both FinTech and Healthcare due to its robust attributes of distribution, decentralisation, permanence and tokenisation (Jain et al., 2021). The authors see data selling as a potential source of financing national health insurance premiums. The DHDM not only leverages the potential of data-backed scientific research but also proposes an achievable solution to unleash the potential of digital health in developing nations such as Africa (Amponsah et al., 2022).

According to Grover et al. (2019) and Grover et al. (2019)'s analysis, Blockchain was widely accepted among social media users attracted by the technology's security, privacy, transparency, trust and traceability aspects.

This paper proposes the design of a medium to high fidelity model of the DHDM (Maher et al., 2020) where patients as data producers will be able to sell their EHR data to consumers (e.g. industrial and pharmaceutical researchers) without compromising their privacy and data security. DHDM envisages providing a solution that will transfer the data ownership to the patient and provide them with the right technological tool to claim their right to control their data under GDPR.EU (2016) chapter 3 Articles 12 - 23 (GDPR EU, 2016) (i.e. patient) provide a framework through which patient can commercialise their data by receiving incentives to encourage sharing and providing technological safeguards to privacy and data protection as well as the implementation of standards to ensure interoperability and quality of data (Figs. 1–3).

2.2. The DHDM data management approach

This research follows a low-to-mid fidelity design approach to prototype the conceptual DHDM model with extensive requirement analysis focusing on health data generation processes, types of data generated, data collection, pre- and post-processing approaches and archiving approaches with emphasis on data security and Blockchain applications for secure EHR sharing of mobile cloud-based e-health systems.

Layer 1: Data production layer. This layer is the patient or data producer end where the health data is being generated. From a technical perspective, it is mainly a layer for integrating and linking the patient EHR from different providers with the DHDM.

Layer 2: Marketplace layer. This is the backbone of the proposed model and can be further divided into three parts. First is the weighing system. This is the internal components that account for the contribution of each data producer. Whenever a data entry is created and added to the patient record, both the patient and the data creator receive an internal token each. For example, suppose a doctor has examined a patient and added a note such as a diagnosis. In that case, two tokens will be generated against the entry, one for the patient as a data owner (subject) and one for the doctor as the data creator (care provider) and added to their respective accounts. Upon data sales transaction, any amount paid by the data consumer shall be distributed among the patients whose data have been sold. The weighing system is directly proportional to the number of tokens in the respective accounts, and whenever a transaction happens, the amount paid will be distributed on the different

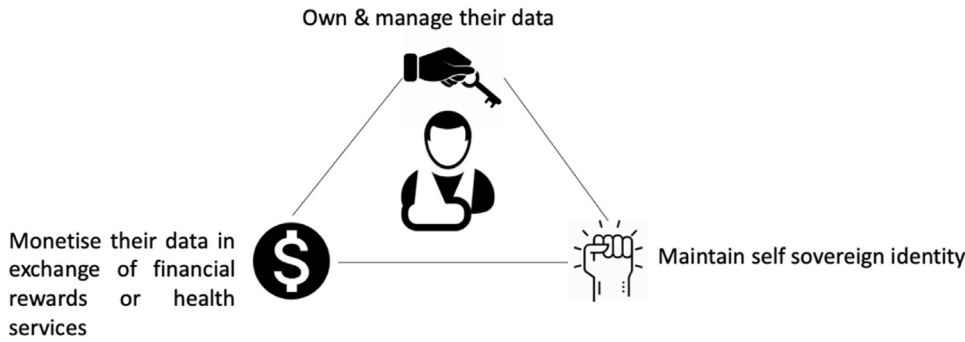


Fig. 1. Three objectives of the DHDM.

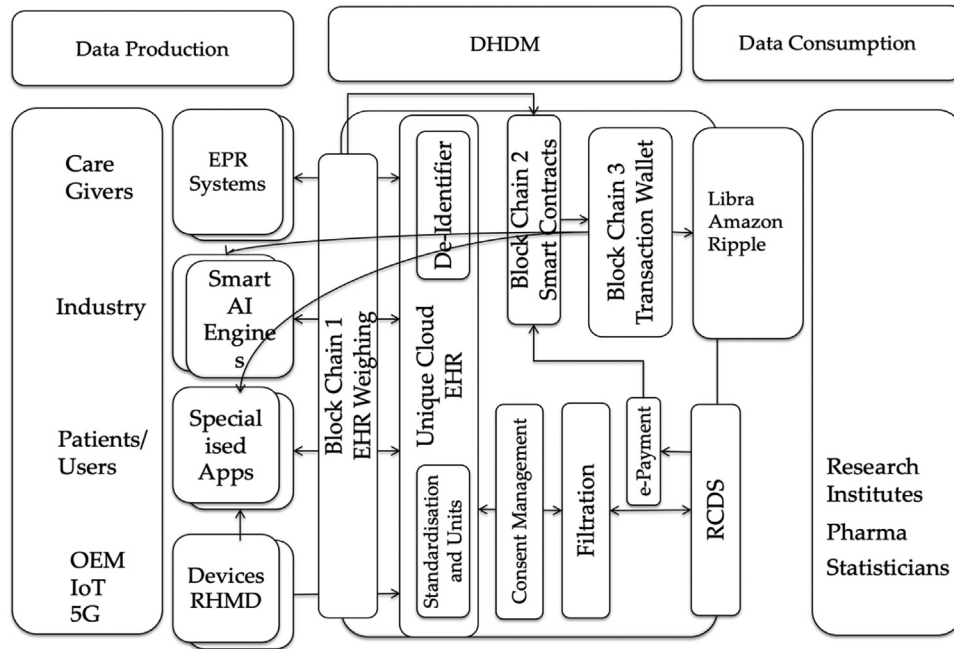


Fig. 2. Mid-fidelity design of DHDM with three interconnected layers – data production, data consumption and marketplace with associated components, technologies, and stakeholders.

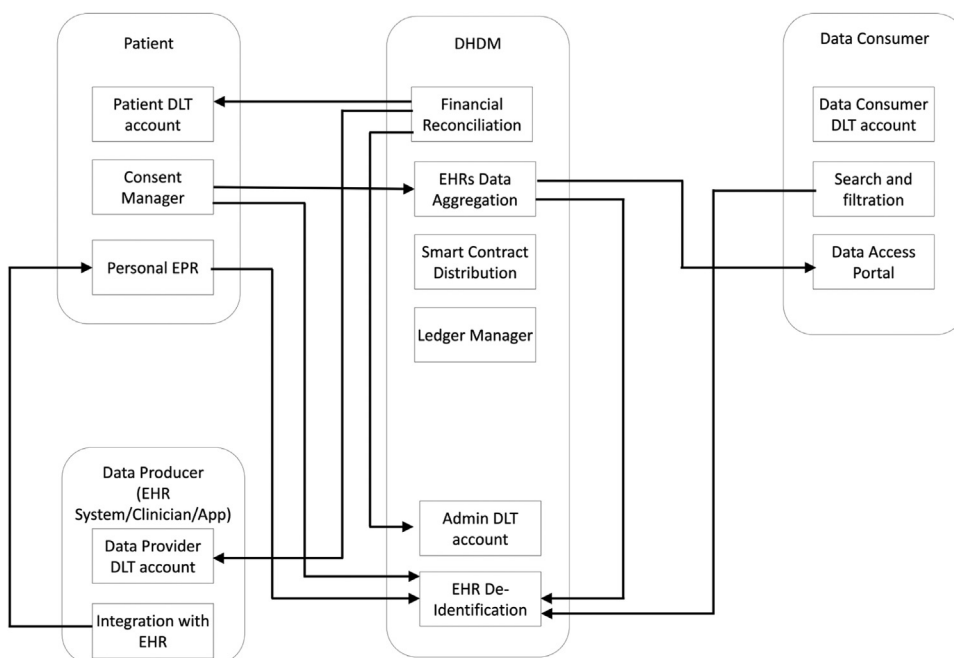


Fig. 3. Operation workflow of DHDM: Four interfaces for the patient, other data producers, researchers and back-end administration (Maher & Khan, 2022).

patients' accounts and split between the patients and their data creators based on the token weight of each account. This guarantees a fair distribution of value between all patients (and their providers) whose records have formed the sold data set.

Second is the EHR Electronic Health Record archive. This is a cloud-based EHR designed to perform a set of functions. It segregates identifiable and de-identifiable data into separate databases. This will help build the anonymised dataset once a transaction happens. It also standardises and unifies the records once received from the data producers. For example, glucose in blood reading in mmol/L would be converted to mg/dL and vice versa. Also, any record coming in nonstandard format will either be standardised or tagged as non-standard, such as Digital Imaging and Communications in Medicine (DICOM) in radiology, Fast Healthcare Interoperability Resources FHIR/HL-7 in text or International Classification of Diseases (ICD) in diagnosis.

Third and final is the smart contract. The smart contract is where data sharing rules, payment distribution, and consent validity will be stored and executed between the data producers and consumers. The value is then transferred to each individual wallet and/or linked to a selected public cryptocurrency.

Layer 3: Data consumption layer. This layer is where the aggregated data sets are presented. It is the researcher or data consumer end. This is where the research search criteria are mapped to a set of data from a group of patient records in the database. This is where the researcher can claim access to the purchased data once the transaction is complete.

The patient has a personal EHR linked to all record sources at different data producers, such as primary care EHR, mobile applications accounts and personal IoT devices. The Patient has a consent manager to manage permissions and finally has a Blockchain wallet account. Other data producers have their own archives related to each patient and their Blockchain wallet account. As a data consumer, the researcher has his Blockchain wallet account, a filtration tool, and a data access portal. DHDM Administration back-end also has a separate Blockchain wallet, a de-identification module, a ledger manager and smart contract.

3. The DHDM stakeholders

Following are the details of the components of the operation process of DHDM:

3.1. Data producers

Under this study, we identified four types of data producers discussed below:

- **Caregivers:** Doctors, therapists, nurses and other medical professionals create entries in the relevant patient records. This includes but is not limited to diagnosis, prognosis, doctor notes, prescriptions and other data components such as pathology and radiology.
- **Patients:** These are the data generated by the patients. It mainly comprises patient-reported experiences (PREMs) and patient-reported outcomes (PROMs).
- **Devices:** Devices are a great source of generating raw data. Such as very simple personal Remote Health Monitoring Devices (RHMDs) such as blood pressure monitors and glucose meters.
- **Industry:** Any Machine Learning engine will be a source of data generation, such as breast cancer screening.

3.2. Data consumers

Three major types of data consumers are identified below:

- **Statisticians and data scientists:** These are researchers looking for aggregated data about a particular population.
- **Researchers:** Especially life science research. Any research analysing medical data to draw findings is a health data consumer.

- **Medical practitioners:** working in health systems where patient records are archived in a fragmented and distributed environment, often find it difficult to make a decision based on the partial view.

3.3. The DHDM workflow

Fig. 4 outlines the transactional workflow of DHDM for both data producer/seller and data consumer/buyer.

- An electronic marketplace where patients share their EHR Electronic Health Records and manage access to them. The more they add data to the EHR, the more valuable their record becomes when it is sold to researchers.
- The Record is then anonymised and put in the proper medical data formats (FHIR, HL-7 and DICOM).
- The units for each parameter are adjusted and placed in interactive graphs.
- Researchers use smart filters to narrow down the data to their research needs.
- They pay for access to the anonymous data.
- The money is distributed between the patients according to the value of each EHR.
- The money is then split between the patient as the data owner and the creator.
- Users are able to buy digital health services from researchers and developers over the platform.

Upon acceptance, the patient will need to link his DHDM account to his/her EPR and, if needed, upload content from a personal archive. The complete EHR is then de-identified by segregating between identifiable and unidentifiable data. In Parallel, a researcher applies his filtration criteria and accesses the EHR database in search of the dataset in question. Once the match is established, a Blockchain Transaction is set where payment is made, and a smart contract distributes the payment between the aggregated EHR owners. At the same time, the researcher is granted access to the data set.

4. SAIL databank background

We identified the Welsh Secure Anonymised Information Linkage (SAIL) Databank ([Secure Anonymised Information Linkage Databank, 2019](#)) as a thriving source of patient data for innovation and policy formulation. The SAIL Databank is the result of intensive scanning of successful data-sharing models all over the world ([WADLB, 2019](#); [Manitoba, Manitoba Centre for Health Policy, 2019](#); [ICES, 2019](#)). SAIL Databank is a nationwide Welsh databank that archives de-identified health datasets gathered from 3 million people in Wales. These datasets are, in turn, made available in an anonymised form to researchers across the world. SAIL databank was established to enable the vast arrays of data collected about individuals during health and other public service delivery to be made available to answer important questions that could not be addressed otherwise.

SAIL is managed and hosted on the UK Secure Research Platform (SeRP) ([Johnson et al., 2020](#); [Jones et al., 2016](#); [Ford et al., 2009](#)). SeRP is an ISO-27001 analytics platform developed by SAIL to control and deliver specific requirements within proper governance. Every request on SeRP is controlled by the National Research Data Appliance (NRDA), a group of modules that form the control management system when SeRP is the analytical interface for researchers to access and perform research.

However, the SAIL databank operation has been disrupted by recent individual rights legislation such as GDPR. The SAIL databank will have to change rapidly to become legitimate under this legislation. This paper highlights the discrepancies in SAIL databank workflow concerning the contradiction of GDPR articles 6 and 9, provisions for a task carried out in the public interest with the GDPR chapter 3: Individual rights

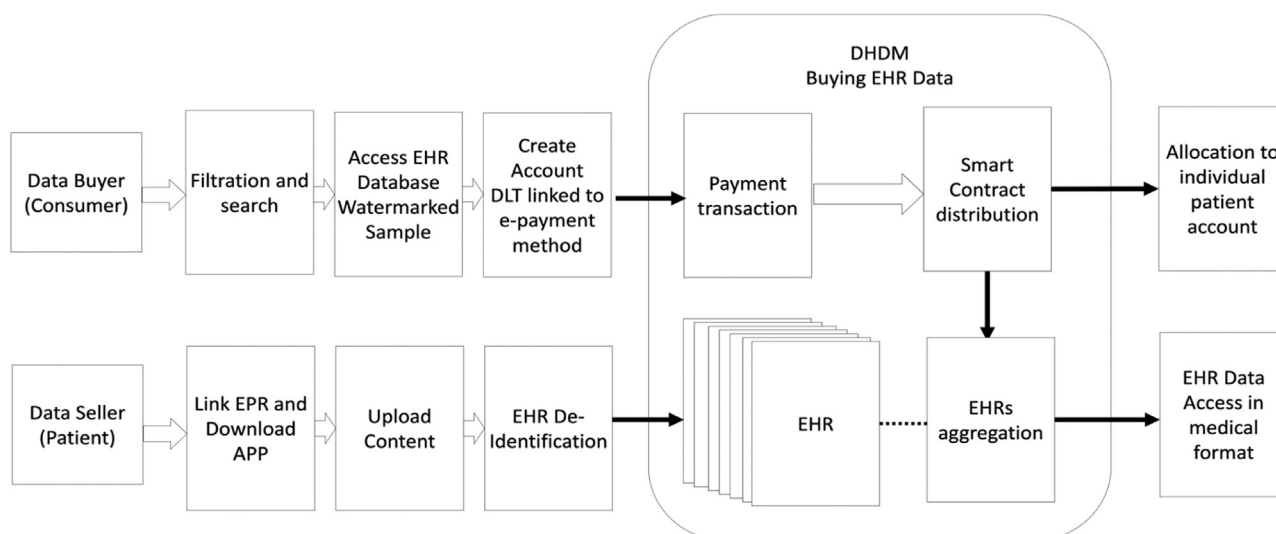


Fig. 4. Transaction Workflow of DHDM.

(GDPR EU, 2016) especially when it comes to Articles 12, 13, 15, 18 and 22.

The article is based on a paradigm shift in the SAIL Databank model by changing the definition of the data provider, from the caregiver to the patient. The DHDM considers the patient as the data owner/provider even when the caregivers created the data. The study aims to validate the assumption that the public notice at the general practitioners (GP) office is sufficient to serve as a consent form. Our understanding of the mentioned articles of the GDPR mandates that the patient gets treated as the data subject and, accordingly, should be given a choice to contribute with their data to every research in which the data will be used. As the data subject, the patient has the right to be informed about each research and opt in or out every time their data meets the inclusion criteria of the research data sample. This includes knowing about the research, its objectives, and the ethical parameters related to the research.

5. Analysis and evaluation of DHDM with SAIL databank

5.1. Analysis of the SAIL databank process workflow

In order to establish the availability of Welsh population health data to researchers, SAIL Databank has taken an aggressive approach toward patient data protection by adopting the so-called privacy by design governance model (Johnson et al., 2020). SAIL Databank created a repository of data gathered from different care providers rather than distributed models. SAIL Databank is subject to compliance with the General Data Protection Regulations GDPR (GDPR EU, 2016), The UK Data Protection Act UK-DPA (DPA, 2018) and the Common Law Duty of Confidentiality CLDC (Council UKCG, 2017).

To evaluate DHDM in conjunction with the SAIL Databank, we first introduce a compliance matrix with four critical aspects of the data marketplace.

- **GDPR Compliance:** The degree to which the process complies with the requirements of the GDPR, specifically Chapter 3: Rights of data subject.
- **Data Flow Efficiency:** The degree seamless transfer of data from one party to another occurs and the importance of the transferred data.
- **Patient Engagement:** The level of involvement of the patients in the process of data creation and in consenting to give access to researchers to use their individual data in their research.
- **Financial Incentives:** The degree of financial incentivisation offered to the stakeholders.

As per the SAIL Databank definition, the data provider is a typical caregiver, NHS, Social Care or other. This positioning treats the caregiver as the data producer and participant in the SAIL Databank. A typical participant is the patient's General Practitioner GP, hospital, outpatient clinic, government department, etc.

The participant (the caregivers in the current SAIL practice) builds their Electronic Health Record (EHR) system in a split database where Patient Identifiable Data (PID) is stored separately from the Clinical Data (CD). PID consists of data that can be used to identify the person to whom the data belongs, such as name, date of birth, address, NHS number, etc., under the GDPR. Anyone holding or processing PID has to justify the need to hold PID.

PID is then passed to the NHS Wales Informatics Services (NWIS), which is an NHS body and therefore has legal rights to hold and process PID. NWIS then creates what they call the Anonymous Linking Field (ALF). The ALF is used to link between PID and CD within the NWIS database. The outcome of the ALF is a joining key that only NWIS can decode and link the key to the PID. The ALF joining Key is sometimes referred to as (File 1) of the patient record, while the CD is referred to as (File 2) of the patient record. NWIS does not hold the CD in their database, only PID and the corresponding ALF joining key (File 1) that is then encrypted and later referred to as (File 3) of the patient record. SAIL Databank then encrypts (File 3) again and combines the double encrypted ALF Key, now called ALF-E, with the CD in a single file that constitutes an entire patient's record in the SAIL database.

On the other end, a data consumer (researcher) who creates a new project must pass the relevant ethics approval depending on their research. The Independent Information Governance Review Panel (IGRP) reviews the project request to access data while complying with information governance, Standard Operating Procedures (SOP), and data management policies.

The SAIL databank operates on a customisable technology and analysis platform, the UK Secure Research Platform (SeRP) (Jones et al., 2016; SERP, 2017). SeRP is where data analysis is processed, and results are concluded based on each research project plan. According to the research requirements, the result is open for viewing by the researcher.

The SAIL Databank had to review its process after the introduction of the GDPR to assess the viability of its consent process. They concluded that they could carry on with their current consent practice by relying on GDPR articles 6 and 9 related to tasks carried out in the public's interest (GDPR EU, 2016), they could carry on with their current consent practice. SAIL's consent process is to provide privacy policy notices in GP

surgeries and similar public waiting areas to inform patients about the data sharing and ask them to come forward if they wish to opt-out. SAIL records show that less than 0.025 per cent of the covered population requested to opt-out of the engagement (Johnson et al., 2020).

6. Case study – evaluation of DHDM in conjunction with the SAIL databank

The DHDM is a framework for a health data marketplace where patients can offer their anonymous electronic health records EHR to researchers and innovators in exchange for incentives, rewards or any other type of compensation. It is a mechanism for collective value co-creation where individual members of the public can contribute to support data-backed research and innovative development of therapeutic and diagnostic systems.

The fundamental change in the proposed process is that the new model suggests that the patient, the GDPR data subject, acts as the data participant and not the caregiver per the current process. i.e. changing the definition of the data provider from the caregivers under the current SAIL Practice to the patient under the DHDM. The objective of the proposed process is to:

- (1) Put access control in the patient's hands by allowing the patient to consent to use their health data in every research proposal where their EHR matches the search criteria and make an informative decision on whether their data is to be included in the research. After reviewing the research proposal and its ethical considerations, they also need controls to grant access to their records. The DHDM access control shall allow the patient to provide on-demand consent when their EHR matches a particular research criterion or by setting rules for their preferred choices. Rules include giving pre-approval for cancer-related research or when the research is conducted by charities yet requests explicit approval for industrial companies such as pharmaceuticals.
- (2) Provide incentives for the patients to participate by requesting incentives against participation in the research. The DHDM proposed that the amounts paid by the researchers to access the data be slotted between the data providers, i.e. patients, while keeping a certain percentage to the SAIL Databank. The aim is to create sustainability for the SAIL Databank operation beyond the government funding currently supporting the operation. It also aims to create wins for the patients to contribute with their data. The DHDM also allows patients to control how they wish to receive the incentives by offering multiple channels such as cash incentives, charity donations, donating to SAIL or receiving rewards in the form of gift vouchers and retail discounts.
- (3) Capture and share the neglected Patient-Reported Data (PRD). Because the current SAIL Databank process considers the caregiver as the data participant, not the patient, some valuable data created by the patient at home are unfortunately neglected. Data such as personal vital signs measurements like glucose levels, blood pressure etc. in the last ten years have seen increased usage of personal devices and lifestyle gadgets that may or may not be medical devices by definition. However, these devices create a wealth of information that is now considered valuable in diagnosis, treatment, and prognosis. They are expected to be as valuable in clinical and industrial research. PRD also includes patient journals, log keeping, Patient Reported Experience Measures PREMs, and Patient-Reported Outcome Measures PROMs. Combining PRD with the professional EHR created by caregivers enriches the value of the Clinical Data CD, which helps equally benefit the patient and the research.
- (4) Create a comprehensive patient-friendly version of the EHR, now named Personal Medical Record (PMR), in the patient's hands to use when receiving services from non-NHS care providers in the UK and overseas. The PMR also allows the patient to contribute to knowledge and awareness creation by sharing parts of their measurements,

experience and records on social-media-like platforms such as PatientsLikeMe (Frost & Massagli, 2008; Wicks et al., 2010) or over traditional non-medical platforms such as WhatsApp.

Fig. 6 presents the proposed flow of information and alteration to the SAIL DataBank process shall they decide to include the patient in their data access system. The process of collecting data from care providers will remain the same. Primary, secondary, and tertiary care systems will still split the EHR into PID (File 1) and CD (File 2). NWIS will continue to create an ALF joining key for File 1 and encrypt it into File 3. SAIL will still combine the double encrypted ALF-E and the CD and make them available for researchers and innovators to search for the data sets they need and request access. However, a new section in the patient EHR will be dedicated to the Patient-Reported Data PRD. PRD will need to be tagged as PRD unless validated by a professional. Researchers accessing the data will decide how reliable the PRD are and decide to include or exclude them in their studies.

7. Discussion

7.1. Contribution to literature

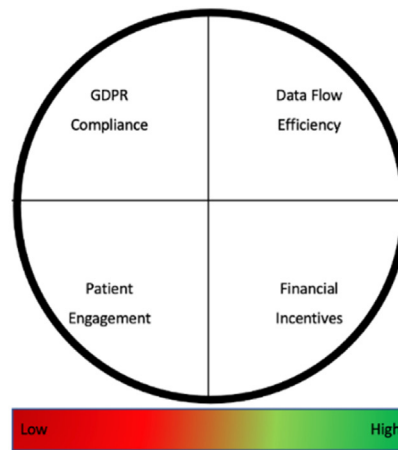
This paper introduced a health data marketplace model that incentivised patients to engage with healthcare systems and the health economy while complying with GDPR. This paper also demonstrated how such a marketplace could be integrated with traditional health data repositories like the Welsh SAIL Databank without perturbing the legacy system's legislative and technical framework. Other databanks like UKBioBank (BioBank UK, UK-biobank, 2022), Health Data Research HDR-UK (BioBank UK, UK-biobank, 2022), Clinical Practice Research Datalink (CPRD) (CPRD, 2022) can also benefit from this model and thereby enable DHDM to be scaled to the national or even international level. However, integration with diverse legacy systems has its technological challenges, particularly from ontological and semantic standardisation and interoperability point of view, which is a pervasive concern in healthcare systems due to its direct impact on patient safety and privacy (Li et al., 2021). Moreover, adopting new and disruptive technology like blockchain is a predicament, particularly in healthcare settings, where personnel involved with data management are reluctant to adopt new technology and resistant to change in data management culture (Martone & Nakamura, 2022). For this type of distributed and diverse environment, the use of Remote Procedure Call (RPC) client within the blockchain-like technologies and native database interface has the demonstrable benefit of integration with minimal perturbation of the legacy system and culture (Shahaab et al., 2021).

Data quality assurance is another crucial concern from the data consumer's point of view. The presentation of the DHDM as a conceptual model for patients to trade their EHR data with researchers and statisticians will increase the majority's appetite to keep updated and complete records of their medical interventions to granular details and with high quality. In the same way the introduction of the credit scoring system will make the population more aware of keeping proper records of their finance, especially with the increased use of wearables and personal medical devices, the type of data generated by the patients is not captured anywhere except databases of the device vendors. It is believed that DHDM adoption will increase the dependencies on EHR combined with patient-reported data from wearables and personal devices. The DHDM has the potential to be implemented on wearables and personal devices data alone, which will label individuals as data producers as well as creators.

7.2. Practical implications

In the paper, The authors discussed the transformation of traditional data repositories in the UK from a government-let opt-out model into a patient-led for-profit opt-in data marketplace and applied the transformation case to the Welsh SAIL DataBank.

A



B

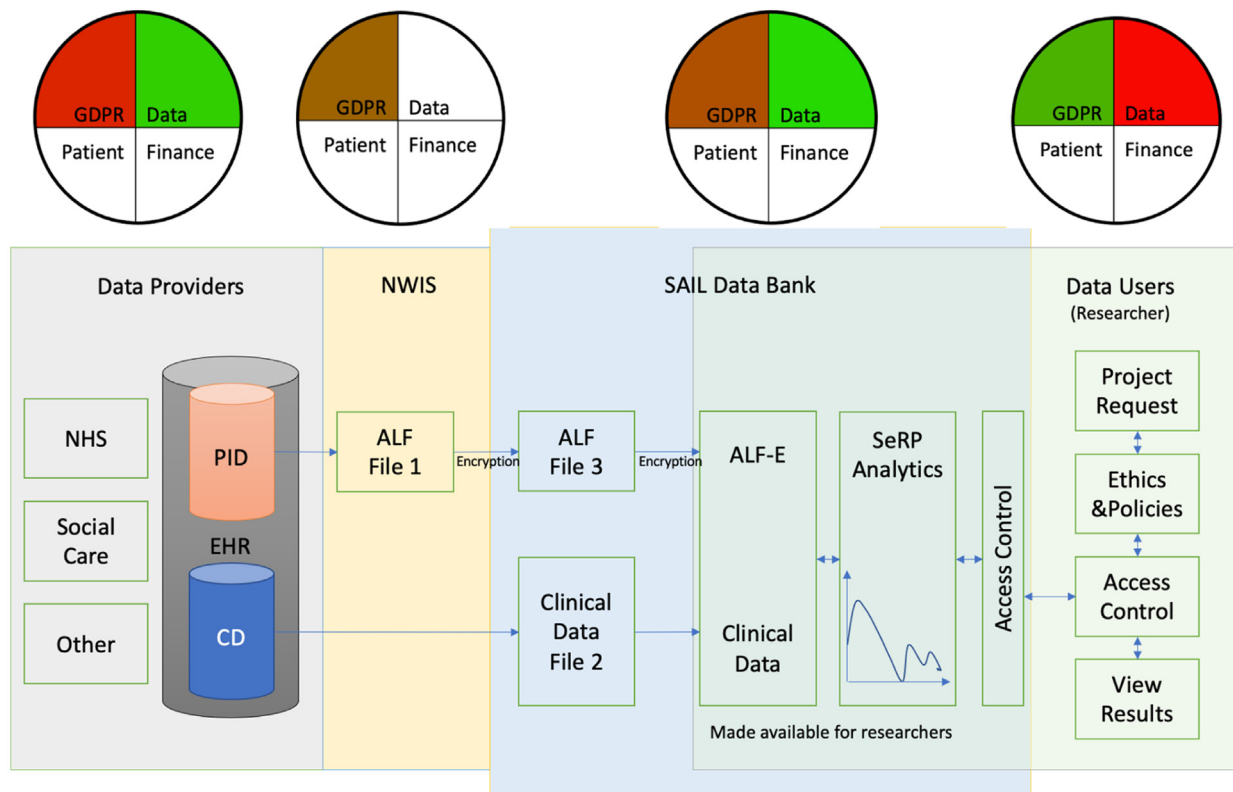


Fig. 5. (A) Compliance matrix to compare the compliance of four aspects of the data marketplace. The colour gradient depicts the degree of compliance. (B) SAIL Databank information flow presents a simplified structure of the privacy by design model of SAIL Databank along with information flow and associated degree of compliance against the compliance matrix.

The significant change in the SAIL process from Figs. 5 to 6 is adding the patient as the data subject in the data access authorisation process. In the new process, when a researcher applies to access some sort of data. All patients identified as potential contributors to the research shall be notified of their engagement interface being online through digital applications or offline through their GPs. This workflow will have to go through the ALF joining key decryption and linking in the NWIS system. SAIL Databank does not have a way to connect patient records and the persona of the record owner except through NWIS. NWIS, on the other hand, although they have the joining key and can identify the data, have no access to clinical data. So, the linkage has to go backwards through the ALF linking and encryption process to notify the patient about the research request.

The patient will be able to access the public description and ethical approval of the research and decide whether they wish to participate. A patient who decides to participate will have to confirm acceptance through a consent process that will tag the patient file as cleared for access by the particular researcher in the ALF linkage and encryption process. This step will also be linked to the reward distribution smart contract blockchain 3 in the DHDM, where the amounts paid by the researchers will be distributed between the data participants, i.e. patients.

The proposed Patient Medical Record (PMR) is a patient-friendly version of the EHR placed in the patient engagement front end. The patient will be free to share the record with other care providers and share parts of the record on social media-like platforms such as PatientsLikeMe. PatientsLikeMe is a social media platform that empowers individuals shar-

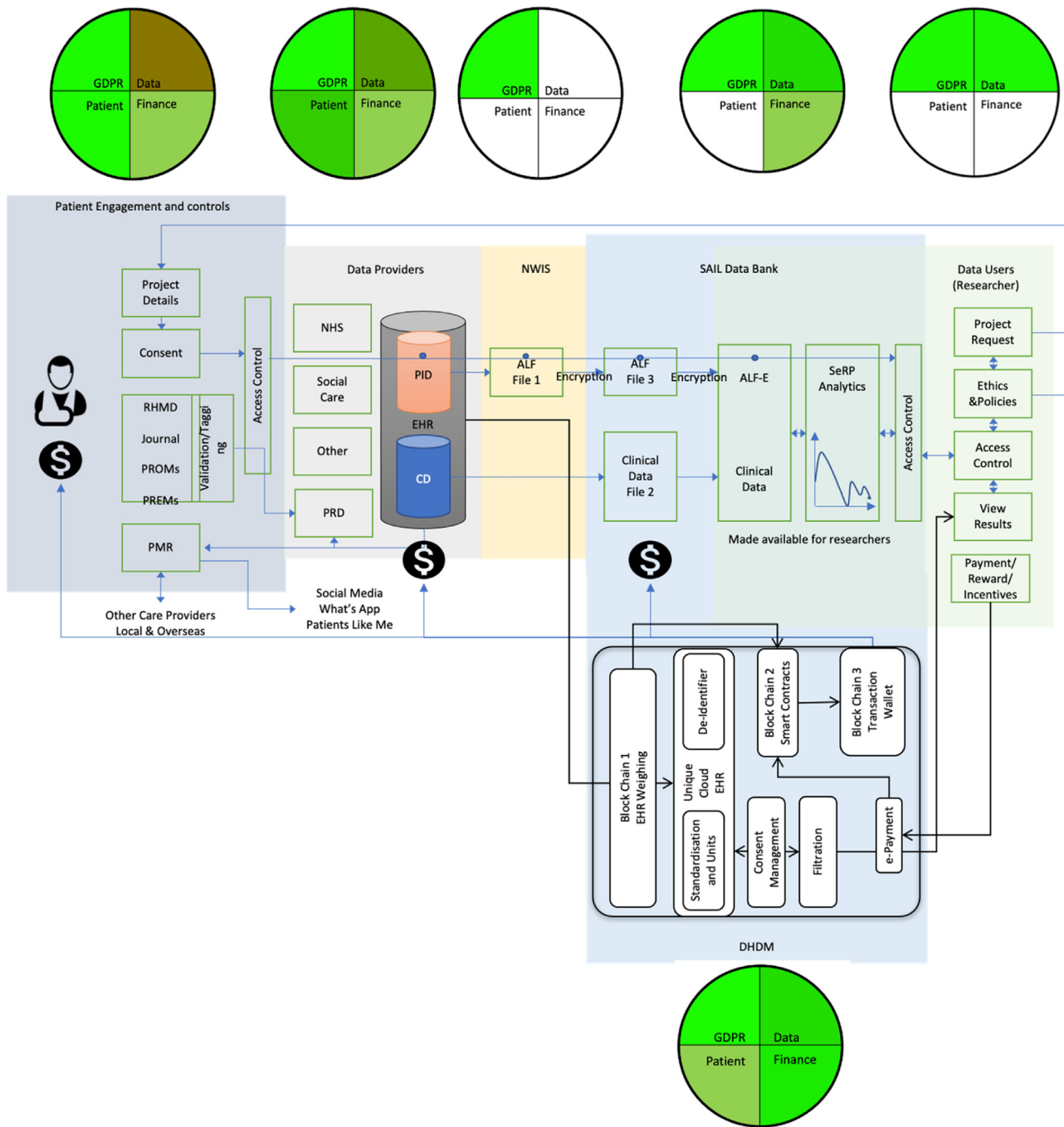


Fig. 6. SAIL Databank integrated DHDM and dataflow compliance matrix.

ing the same health concerns to exchange views and experiences around common interests. Studies showed a growing demand for patient access to personal health records (Wicks et al., 2010). A field survey has established that members of the PatientsLikeMe communities perceive various benefits (Frost & Massagli, 2008). Therefore, their experience needs to be closely monitored and evaluated is an essential step in building a comprehensive PMR and availing PRD for researchers.

The PMR is proposed to take a Human Medico-Social approach that consolidates the medical data in one place to be readable and accessible by non-professionals. The PMR increases access to health services & reach, increases awareness of health concerns, improves caregivers' knowledge & capacity to deliver better service, Improves the participa-

tion of concerned individuals & groups in uplifting health services and provides the base for enhancing & adding services.

8. Conclusion

The increase in EHR adoption will, furthermore, positively affect longterm patient outcomes. The direct positive impact mainly comes from early detection, prevention, and improved self-management. From an information systems perspective, the DHDM introduced a data-backed research and evidence-based framework that can benefit machine learning algorithms and diagnostics AI engines. Such development will support value creation that should lead to better pre-infestation

alarming/alert in early detection, Computer Aided Diagnosis (CAD) and evidence-based prognosis. In the scope of our research, the researcher intends to design comprehensive correlational research as a future build-up on our work where future researchers will be able to test and understand the dependencies and causality of the patient's digital engagement with the health system and their medium to long terms expected clinical outcomes.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ijmei.2023.100159.

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